Hesai AT512 for Emergency Brake Assist

Lidar performance – statistical modelling (Exercise 3)

Team: *names TBD*

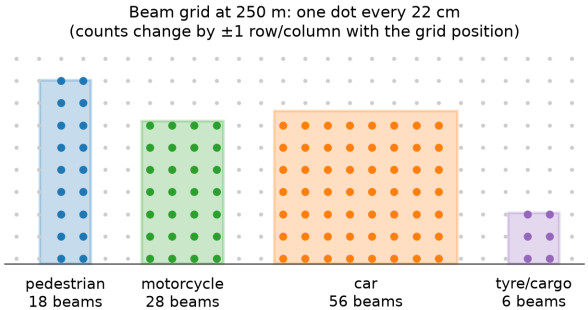
3DSSF – Kozma

The lidar: Hesai AT512

- 300 m @ 10 % reflectivity (400 m max)
- 0.05° × 0.05° resolution 2400 × 512 points @ 10 Hz
- 120° × 25.6° field of view → covers adjacent lanes
- automotive-grade series product

Fine grid ⇒ many beams on small, far objects

Assumption: the 300 m spec means 90 % detection probability (not stated by Hesai).



Model: returns per frame are a random variable

1. Beams on target (W × H at R)

$$n = \frac{W}{R\,\Delta\theta_h} \cdot \frac{H}{R\,\Delta\theta_v}$$

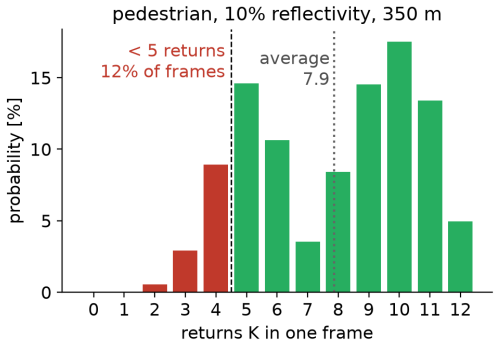
random grid offset → n rounds up or down

2. Each beam returns with prob.

$$p = 1 - e^{-C\rho/R^2}, \quad p(300\,\text{m}, 10\,\%) = 0.9$$

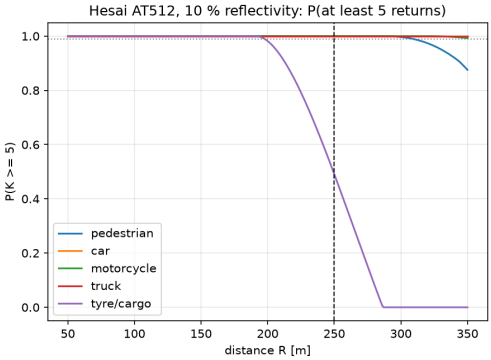
3. Returns: $K \sim \text{Binomial}(n, p)$

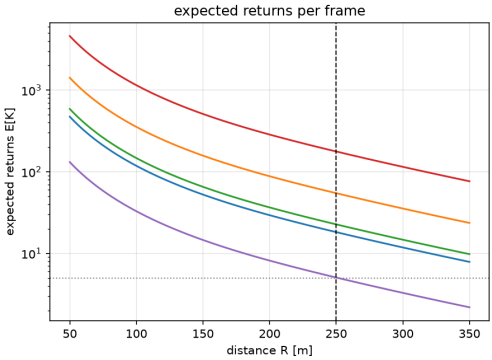
Requirement: $P(K \geq 5) \geq 99\,\%$ at 250 m



average $7.9 \geq 5$, but 12 % of frames fail
⇒ **the average passes, the probability fails**

Results (10 % reflectivity)





$P(K \geq 5)$ at 250 m: pedestrian and all vehicles $> 99.99\,\%$, tyre (0.5 m) **49 %**

Conclusion: the AT512 fulfils EBA

- all vehicles and pedestrians pass at 250 m, even at 10 % reflectivity
- small debris ($\sim$0.5 m): only up to $\approx$ 200 m
  too few beams hit – a **resolution** limit, not reflectivity

Limitations: flat frontal targets; independent beams (optimistic in rain / dirty sensor); 90 % behind the range spec assumed; single frame – 34 more frames before braking must start.

